Blockchain Applications

Course syllabus · Academic year 2026–2027, semester I · Version of 7 November 2026

1. Course identification

Item	Detail
Programme	Master Blockchain for Business (MBB), year I, semester I, 2026–2027
Faculty	Faculty of Business Administration in Foreign Languages (FABIZ), Bucharest University of Economic Studies
Language	English (the course website is also available in Romanian)
Structure	One course in two parts, one final grade
Part 1: Crypto Economics	Prof. Ilan Alon, Ariel University and Corvinus University of Budapest, ilan.alon@uni-corvinus.hu. Monday 16 – Saturday 21 November 2026; weekdays 17:30–20:40, Saturday 09:10–15:40 (28 academic hours); Room 4113, 1st floor (Trading Lab), Victor Slăvescu Building, 2–2A Calea Griviței, District 1, Bucharest
Part 2: Blockchain Analytics	Prof. dr. Daniel Traian Pele, Department of Statistics and Econometrics, Bucharest University of Economic Studies, and Institute for Economic Forecasting, Romanian Academy, danpele@ase.ro. 8 lectures: Mon 23 Nov (L1–L2), Tue 24 Nov (L3–L4), Wed 25 Nov (L5–L6), Thu 26 Nov (L7–L8); Room 4113, 1st floor (Trading Lab), Victor Slăvescu Building, 2–2A Calea Griviței, District 1, Bucharest
Exam	Friday 27 November 2026, 60 minutes, online on the course website
Course website	https://danpele.github.io/Blockchain-Applications/

2. Course description

Blockchain Applications is one course in two parts that look at the same applications (Bitcoin, Ethereum, stablecoins, tokens, DAOs and DeFi) from two sides.

- **Part 1, Crypto Economics** (Prof. Ilan Alon), asks how crypto applications create and capture value. It works through five business cases, the Bitcoin and Ethereum whitepapers and academic articles, and builds a due-diligence project in teams, with one hour of guided project work every weekday.
- **Part 2, Blockchain Analytics** (Prof. Daniel Traian Pele), treats the same applications as data generators. Every transaction, fee and transfer is public, time-stamped and free to analyse; eight lectures turn that data into statistical evidence: how risky an asset is, whether a network is growing, when a stablecoin is under stress, who moves together with whom, and which forecasts survive an honest out-of-sample test.

The course builds on what students already know and does not repeat it:

- **Introduction to Blockchain Technology** (October 2026) covered the technical foundations, so both parts start from use cases, not from hashing and consensus.
- **Blockchain Business Models 1** (October–November 2026) covered business-model design; Part 1 applies it to crypto cases and Part 2 adds the evidence from the data.
- **Blockchain Legal Framework** (December 2026) and **Finance and Blockchain** (January–February 2027) come later, so regulation enters only where a use case depends on it, and capital-markets finance is left to that course.

Part 2 is quantitative, at master level. Each lecture starts from a real application and treats it with a formal statistical model: the model, the estimator, its standard error and the test statistic under the null hypothesis. Every number in the lectures is computed from real data, and students reproduce it in Google Colab notebooks.

Prerequisites. Introductory statistics and econometrics (regression, hypothesis testing). Basic Python helps; the notebooks provide working code and run in the browser.

3. Learning outcomes

By the end of the course, students will be able to:

Part 1: Crypto Economics

1. **Explain** blockchain and Web3, and judge when a blockchain is worth its cost.
2. **Assess** smart contracts and decentralised governance.
3. **Analyse** token incentives and value capture.
4. **Evaluate** crypto business models and digital assets.
5. **Defend** an evidence-based due-diligence recommendation.

Part 2: Blockchain Analytics

6. **Collect** market and on-chain data from public sources (EODHD, DefiLlama, Coin Metrics, block explorers) and check its quality before any analysis.
7. **Describe** crypto returns with the right statistics and test their stylized facts (heavy tails, volatility clustering, changing correlations) against stocks and gold.
8. **Test** market efficiency, date bubbles and crashes with drawdowns and the LPPL model, and judge network-value claims such as Metcalfe's law.
9. **Construct** a crypto market index (CRIX, with the number of constituents chosen by the AIC) and a crypto volatility index (VCRIX), and explain how an index becomes an investible fund.
10. **Measure** risk with EWMA and GARCH volatility, Value-at-Risk (VaR) and Expected Shortfall (ES), and backtest the forecasts with the Kupiec and Christoffersen tests and the Basel traffic light.
11. **Monitor** stablecoin depeg risk with peg deviations, AMM price impact and on-chain early-warning signals, and evaluate the warnings event by event.
12. **Estimate** systemic depegging pressure with quantile regression and the Quantile Lasso, build the De-pegging Pressure Index (DPI) and test it around stress events.
13. **Map** crypto markets as networks, classify crypto assets against traditional asset classes, and measure the effect of protocol upgrades and sentiment with event studies.
14. **Evaluate** machine-learning and AI forecasts out of sample: detect overfitting and data snooping, and compare risk forecasts with proper scoring rules and Diebold–Mariano tests.

4. Course materials

All materials are on the course website, https://danpele.github.io/Blockchain-Applications/.

Part 1: Crypto Economics

For each day: the whitepaper or academic article to read, the business case, short videos, and the project milestone for the guided project hour (section 5). The five cases (Ivey, The CASE Journal, The Case Centre) are provided as PDF copies for enrolled students: the links on the course website open them with the ASE Google account.

Part 2: Blockchain Analytics

For each lecture:

- **Slides** (PDF), with clickable references. Each deck opens with a quotation and ends with an appendix of further material for self-study.
- **Web page**: the lecture written as continuous text, like a textbook chapter, with every chart and worked example of the slides.
- **Interactive page**: the key charts with parameters students can change, and questions for the room.
- **Colab notebook**: reproduces every number and chart of the lecture from the course data; it needs only a Google account.
- **Quantlets**: one short notebook per chart, in Quantlet format, on GitHub.
- **Practice quiz**: 15 questions drawn at random from a bank of 30 per lecture, with immediate feedback. It does not count towards the grade and no score is stored.
- **Video units** of the Blockchain Analytics video course on Quantinar, where a lecture has matching units.

Data. Daily and hourly prices of cryptos, stablecoins, stocks, indices, FX and gold come from EODHD and are stored in the course repository, so the notebooks need no data subscription or API key. The lectures use daily data up to 23 October 2026. Gold, the S&P 500 and FX use Monday–Friday closes (252 trading days a year); cryptos trade every day (365 days a year).

5. Part 1 schedule: Crypto Economics (Prof. Ilan Alon)

Weekdays: 17:30–19:00 and 19:10–20:40, with guided project work from 19:40 to 20:40 (the instructor advises each team). Saturday: team presentations in the classroom, 09:10–15:40 (four 90-minute blocks), 20 minutes per team plus 10 minutes of questions. Prepare the reading, case or whitepaper before each day. All times are Bucharest time.

Day	Focus	Readings and case	Project hour
Mon 16 Nov	**Bitcoin and its whitepaper** Blockchain foundations, the whitepaper and macroeconomics; demand, supply, network effects, costs and value of Bitcoin	Nakamoto (2008), Bitcoin whitepaper (in full, before class) Jiang, Rodríguez & Zhang (2023), Financial Management (PDF, ASE account): introduction and conclusion Case: Cryptocurrencies: Investment, Money, or Gamble? (A) (The Case Centre) (PDF copy, ASE account)	Form teams and select the project; define its use case and economic value proposition
Tue 17 Nov	**Ethereum and smart contracts** Accounts, state transitions, gas; decentralised applications, NFTs, execution and security risks	Ethereum whitepaper Singh, Gaur & Singh (2024), Blockchain-based governance, British J. Management (PDF) Case: Polygon Labs: Scaling Web3 Growth with Cost per Wallet Efficiency (Ivey) (PDF copy, ASE account)	Map the blockchain architecture and smart-contract functions, dependencies and risks
Wed 18 Nov	**Financial markets and tokenomics** Stablecoins, CBDCs, wallets, custody; supply, unlocks, incentives and value capture; the XRP case	Kampakis (2018), Three case studies in tokenomics, J. British Blockchain Association (PDF) Case: The ripple effect: navigating an XRP investment (The CASE Journal) (PDF copy, ASE account)	Token supply and allocation table; dilution and custody risks; bull and bear scenarios
Thu 19 Nov	**DAOs and decentralised governance** Code, governance and accountability; voting, treasury control, delegation, stakeholder incentives; ethics and sustainability	Alon, Klemetsen, Šilenskytė & Gildin (2025), New pathways for international business governance via blockchain-based DAOs, Multinational Business Review Case: The DAO Hack: A Blockchain Dilemma (Ivey) (PDF copy, ASE account)	Map governance and the business model; security, accountability and stakeholder risks
Fri 20 Nov	**DeFi, valuation and liquidity** DEXs, AMMs, lending, collateral and liquidation; token versus equity; market cap, FDV, comparables, scenarios, slippage and exit	Piñeiro-Chousa et al. (2022), DeFi cryptocurrencies and other assets, Technological Forecasting and Social Change (PDF) Case: KaspaCom: Capturing Value from the Kaspa Blockchain (Ivey) (PDF copy, ASE account)	Valuation comparables; bear, base and bull scenarios; liquidity and exit risk
Sat 21 Nov	**Due-diligence presentations** Team presentations, questions and comparative feedback; course synthesis	Lo & Medda (2020), Assets on the blockchain: an empirical study of tokenomics, Information Economics and Policy (PDF, ASE account) The whitepaper or technical documentation of the team's project	–

Videos for each day (Prof. Alon and Coin Bureau) are linked on the course website.

6. Part 2 schedule: Blockchain Analytics (Prof. Daniel Traian Pele)

Each lecture pairs one application with one statistical method and one Colab notebook. Working papers are cited as such: they are not yet published.

Lecture	Topic and sections	Statistical methods	Data and cases	Core reading	Quantinar units
L1 Mon 23 Nov	**Blockchains and Crypto Markets as Data** *What a blockchain records; the market in numbers; on-chain activity; data quality; growth on a log scale*	Proof-of-work arithmetic and the supply rule; market concentration (Herfindahl–Hirschman index, HHI); data-quality checks on prices and on-chain series; log-linear trends with Newey–West standard errors	Bitcoin and Ethereum prices and market values (EODHD); on-chain activity (Coin Metrics, Ethereum on-chain data); mining energy (Cambridge CBECI); stablecoin supply (DefiLlama)	Härdle, Harvey & Reule (2020), Understanding cryptocurrencies, J. Financial Econometrics Makarov & Schoar (2020), Trading and arbitrage in cryptocurrency markets, J. Financial Economics	Cryptos, NFTs, Digital Assets; Blockchain Technology – An Introduction; Understanding Cryptocurrencies
L2 Mon 23 Nov	**Stylized Facts, Efficiency and Bubbles** *Prices and returns; the shape of returns; volatility and correlation; market efficiency; bubbles, crashes and network value*	Moments and the Jarque–Bera test; ACF and Ljung–Box; rolling correlations; variance-ratio tests; drawdowns and the log-periodic power law (LPPL) model; Metcalfe's law	Daily returns of Bitcoin, Ethereum, the S&P 500 and gold (EODHD); the 2017 Bitcoin bubble	Trimborn & Härdle (2018), CRIX: an index for cryptocurrencies, J. Empirical Finance Urquhart (2016), The inefficiency of Bitcoin, Economics Letters Peterson (2018), Metcalfe's law as a model for Bitcoin's value	Blockchain Mechanism and Distributional Characteristics of Cryptos; Crypto Currency Returns
L3 Tue 24 Nov	**Crypto Indices: CRIX, VCRIX and Investible Index Funds** *Motivation for a crypto index; constructing CRIX; replication on real data; VCRIX, a volatility index; investible index products; nine types of crypto indices*	Index construction (Laspeyres index, divisor, choice of the number of constituents by the Akaike information criterion, AIC); the CRIX family and the VCRIX volatility index; nine types of crypto indices (market, sector, price, futures, volatility, yield, smart beta, fundamentals, sentiment)	The CRIX family of indices and VCRIX; from index to investible fund (Royalton CRIX, a spot Bitcoin ETF); square-root weights and a protection floor on real data	Trimborn & Härdle (2018), CRIX an index for cryptocurrencies, J. Empirical Finance Kim, Trimborn & Härdle (2021), VCRIX — a volatility index for crypto-currencies, International Review of Financial Analysis Găman, Będowska-Sójka, Pele & Härdle (2026), White paper on cryptocurrency indices	—
L4 Tue 24 Nov	**Volatility and Tail Risk** *Volatility models; estimates; VaR and ES; backtesting; out of sample*	EWMA and GARCH(1,1)-t by maximum likelihood; residual diagnostics; VaR 1% and ES 2.5% (Normal, historical, EWMA, GARCH-t); Kupiec and Christoffersen backtests; Basel traffic light	Bitcoin daily returns: estimation up to 18 September 2026, out-of-sample check from 19 September to 23 October 2026	Bollerslev (1986), Generalized autoregressive conditional heteroskedasticity, J. Econometrics Christoffersen (1998), Evaluating interval forecasts, International Economic Review	Robust Markowitz for Crypto-Assets
L5 Wed 25 Nov	**Stablecoins I: Measuring Depeg Risk and On-Chain Early Warning** *What a stablecoin promises; measuring the peg; automated market makers (AMMs) and price formation; early warning; recent data, August to October 2026*	Peg deviations in basis points; exceedance curves and episode counts; AMM price impact (constant product, StableSwap); event-level evaluation of early-warning signals	Daily and hourly USDT, USDC and DAI prices up to 23 October 2026 (EODHD); hourly DeFi pool data (Chaffard, 2026, Zenodo); case: the March 2023 USDC depeg	Lyons & Viswanath-Natraj (2023), What keeps stablecoins stable?, J. International Money and Finance Chaffard, Găman, Staures, Ginavar, Jheng, Härdle & Pele (2026), Onchain monitoring of stablecoin depeg risk, working paper Zouari, Alon & Shtudiner (2026), How fast can a run be? Crisis speed across stablecoins, banks, and money market funds, J. Economics and Business	—

Lecture	Topic and sections	Statistical methods	Data and cases	Core reading	Quantinar units
L6 Wed 25 Nov	**Stablecoins II: Systemic Depegging Pressure and Portfolios** *Comparing different anchors; quantile regression and the Lasso; the De-pegging Pressure Index; tail dependence between stablecoins*	Reference-adjusted deviations; check loss and quantile regression; 5% Quantile Lasso with GACV selection; the De-pegging Pressure Index (DPI); HAC event tests with Holm correction; out-of-sample check-loss skill	Daily prices of 10 stablecoins up to 23 October 2026 (EODHD); stress episodes from FTX (2022) and Silicon Valley Bank (2023) to BUSD (2025)	Koenker & Bassett (1978), Regression quantiles, Econometrica Fan, Sha, Härdle & Pele (2026), Reference-adjusted Quantile Lasso reveals regime-dependent stablecoin depegging pressure, working paper	FRM for Cryptos
L7 Thu 26 Nov	**Networks, Sentiment and Asset Classes** *Crypto markets as networks; cryptocurrencies as an asset class; event studies of Ethereum upgrades; sentiment from social media*	Correlation distance and minimum spanning trees; absorption ratio; Diebold–Yilmaz spillovers; factor analysis and k-means classification of asset classes; market-model event study (CAR, cross-sectional tests); text sentiment	Daily returns of 33 assets (19 coins and 14 traditional assets); 906 assets for the classification (234 cryptos); Ethereum upgrades from London to Dencun; BitMood sentiment	Pele et al. (2023), Are cryptos becoming alternative assets?, European J. Finance Diebold & Yilmaz (2012), Better to give than to receive, Int. J. Forecasting Lin, Găman, Wang & Pele (2026), When does the market react to protocol innovation?, working paper	Dynamic Crypto Networks; Cross-exchange Crypto Risk; Reddit Sentiment and BTC Returns; Elon Musk's Twitter Feed; DAI Digital Art Index
L8 Thu 26 Nov	**Machine Learning, AI and Their Limits; Exam Review** *Machine-learning forecasts of Bitcoin's direction; data snooping; risk forecasts (classical, ML, pretrained); the limits of ML and AI; exam review*	Walk-forward evaluation; Brier score, AUC and the Pesaran–Timmermann test; overfitting and SHAP explanations; the deflated Sharpe ratio; VaR from GARCH, quantile regression, gradient-boosted trees and the pretrained Chronos-2 model; quantile score and Diebold–Mariano tests; look-ahead in pretrained models	Bitcoin daily prices up to 23 October 2026 (EODHD); zero-shot Chronos-2 forecasts	Bailey & López de Prado (2014), The deflated Sharpe ratio, J. Portfolio Management Gneiting & Resin (2023), Regression diagnostics meets forecast evaluation, Electronic J. Statistics Ansari et al. (2025), Chronos-2, arXiv	AI Meets Blockchain; Crypto-Backed P2P Lending

7. Assessment

Component	Weight	Format
Due-diligence project (Part 1, Prof. Alon)	50%	Team slides, source log and a 20-minute presentation in the classroom plus 10 minutes of questions, Saturday 21 November 2026
Exam (both parts, one paper), Friday 27 November 2026	50%	60 minutes, on the course website; 30 multiple-choice questions; no books, notes or other materials

Due-diligence project (Prof. Alon)

Five teams of four students. Each team takes a different category (AI; gaming and game assets; decentralized storage and cloud; DeFi lending and decentralized exchanges; on-chain trading and derivatives; tokenized real-world assets; privacy; oracles and interoperability; payments and monetary networks; exchange-linked tokens; meme and community-driven assets) and one cryptocurrency from it, so no two teams share a category or a coin; the candidate coins are listed in the assignment and on the course website. Chosen on Monday 16 November. The memo answers one decision question: should an informed investor or strategic partner Buy, Hold, Avoid or Watch the asset? For a project without an investable token the answer is Partner, Do Not Partner or Watch. Teams must separate project quality from token investment quality: a useful project can have a poor token.

- **Deliverables:** 10–12 slides and a source log; a 20-minute boardroom-style presentation in the classroom on Saturday 21 November, followed by 10 minutes of questions; the presentation is also uploaded to YouTube.
- **Required exhibits:** project snapshot, market map, token unlock chart, value-capture diagram, valuation comparables, governance map, risk heatmap and final recommendation.
- **Coverage:** problem and use case; technology and competition; business model and tokenomics; evidence and valuation (market capitalisation, FDV, peers); governance and risk; bull and bear cases with the evidence that would change the recommendation.
- **Criteria, 25% each:** investment thesis and crypto-economic analysis; research and analytical rigour; critical thinking and risk assessment; presentation quality and engagement.

Assignment and student template: course website. The project is an academic exercise, not financial advice. Cite sources and data dates, separate facts from assumptions, and disclose substantive AI assistance.

Exam

One exam, taken on the course website: students sign in with their ASE Google account and type the exam code shown on the screen. It has 30 multiple-choice questions, one point each: 15 on Crypto Economics and 15 on Blockchain Analytics, mixed together. The last 5 questions are about a short case built on real output from the course notebooks. The score is the share of correct answers, out of 100.

Each student gets a different set of questions on the screen: they are drawn from larger question banks, with several variants of each question and case, and the order of the questions and of their options is shuffled. No calculations are required, and a wrong answer costs nothing. Answers are saved on the server while the student works; if the page closes, the student signs in again and continues with the same deadline. Leaving the exam page is recorded. The practice quizzes and a practice exam help to prepare.

8. Readings, data and tools

Part 1: Crypto Economics

The whitepapers, articles and cases of each day are listed in section 5, with links.

Part 2: Blockchain Analytics, core texts

- Bailey, D. H., & López de Prado, M. (2014). The deflated Sharpe ratio: correcting for selection bias, backtest overfitting, and non-normality. *The Journal of Portfolio Management*, 40(5), 94–107.
- Bollerslev, T. (1986). Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 31(3), 307–327.
- Christoffersen, P. F. (1998). Evaluating interval forecasts. *International Economic Review*, 39(4), 841–862.
- Diebold, F. X., & Yilmaz, K. (2012). Better to give than to receive: predictive directional measurement of volatility spillovers. *International Journal of Forecasting*, 28(1), 57–66.
- Găman, Ș., Będowska-Sójka, B., Pele, D. T., & Härdle, W. K. (2026). White paper on cryptocurrency indices. IDA Institute Digital Assets, working paper.
- Gneiting, T., & Resin, J. (2023). Regression diagnostics meets forecast evaluation: conditional calibration, reliability diagrams, and coefficient of determination. *Electronic Journal of Statistics*, 17(2), 3226–3286.
- Härdle, W. K. (2026). System and method for managing an investible cryptocurrency index fund. US Patent 12,784,107 B2, granted 3 November 2026; assignee Royalton Partners AG.
- Härdle, W. K., Harvey, C. R., & Reule, R. C. G. (2020). Understanding cryptocurrencies. *Journal of Financial Econometrics*, 18(2), 181–208.
- Kim, A., Trimborn, S., & Härdle, W. K. (2021). VCRIX — a volatility index for crypto-currencies. *International Review of Financial Analysis*, 78, 101915.
- Koenker, R., & Bassett, G. (1978). Regression quantiles. *Econometrica*, 46(1), 33–50.
- Lyons, R. K., & Viswanath-Natraj, G. (2023). What keeps stablecoins stable? *Journal of International Money and Finance*, 131, 102777.
- Makarov, I., & Schoar, A. (2020). Trading and arbitrage in cryptocurrency markets. *Journal of Financial Economics*, 135(2), 293–319.
- Pele, D. T., Wesselhöfft, N., Härdle, W. K., Kolossiatis, M., & Yatracos, Y. G. (2023). Are cryptos becoming alternative assets? *The European Journal of Finance*, 29(10), 1064–1105.
- Trimborn, S., & Härdle, W. K. (2018). CRIX an index for cryptocurrencies. *Journal of Empirical Finance*, 49, 107–122.
- Urquhart, A. (2016). The inefficiency of Bitcoin. *Economics Letters*, 148, 80–82.
- Zouari, M., Alon, I., & Shtudiner, Z. (2026). How fast can a run be? Crisis speed across stablecoins, banks, and money market funds. *Journal of Economics and Business*, 106344.

Each slide deck lists its full references, with links.

Video courses

Part 1: short videos by Prof. Alon and Coin Bureau, linked on the course website for each day. Part 2: Blockchain Analytics on Quantinar, 23 units with slides and recorded lectures.

Data sources

- EODHD: daily and hourly prices of cryptos, stablecoins, stocks, indices, FX and gold (stored in the course repository).
- DefiLlama: stablecoin supply and DeFi total value locked (TVL).
- Coin Metrics: on-chain activity of Bitcoin and Ethereum (community data).
- Cambridge Bitcoin Electricity Consumption Index (CBECI): mining energy and the geography of mining.
- Further public sources, useful for the project: CoinGecko, Etherscan, Dune, DefiLlama.

Tools

Google Colab notebooks in Python, run in the browser; students need only a Google account. Code and data: github.com/danpele/Blockchain-Applications.

9. Course policies

- **Attendance.** Attendance is recorded at every lecture of both parts: students sign in on the course website with their ASE Google account and type the code shown on the screen, which changes every minute. Attendance does not count towards the grade, but attending every lecture is strongly recommended.
- **AI use.** Students may use AI tools to study, to run the notebooks and for the project, and disclose substantive AI assistance. AI tools are not allowed during the exam.
- **Contact.** Prof. Ilan Alon (Part 1 and the project), ilan.alon@uni-corvinus.hu; Prof. Daniel Traian Pele (Part 2 and the exam), danpele@ase.ro. Questions are answered by e-mail, or before and after class.